

## Sensor Modality Selection – Range

Scenario: Lane Change in front of the Ego car on US101 Highway

| Trajectory Prediction Model |                        | CS-LSTM             |      |      |      |      |                     |      |      |      |      | STDAN               |       |      |      |       |                     |       |       |       |       |
|-----------------------------|------------------------|---------------------|------|------|------|------|---------------------|------|------|------|------|---------------------|-------|------|------|-------|---------------------|-------|-------|-------|-------|
| Manouver                    |                        | Right Lane change   |      |      |      |      | Left Lane change    |      |      |      |      | Right Lane change   |       |      |      |       | Left Lane change    |       |       |       |       |
| Sensor Parameter            |                        | RMSE (m) at 'Range' |      |      |      |      | RMSE (m) at 'Range' |      |      |      |      | RMSE (m) at 'Range' |       |      |      |       | RMSE (m) at 'Range' |       |       |       |       |
| Sensor Modality             | Prediction Horizon (s) | 50m                 | 75m  | 100m | 125m | 150m | 50m                 | 75m  | 100m | 125m | 150m | 50m                 | 75m   | 100m | 125m | 150m  | 50m                 | 75m   | 100m  | 125m  | 150m  |
| Radar                       | 1                      | 0.38                | 0.36 | 0.37 | 0.37 | 0.36 | 0.73                | 0.7  | 0.54 | 0.67 | 0.63 | 0.52                | 0.52  | 0.52 | 0.52 | 0.53  | 0.84                | 0.83  | 0.81  | 0.84  | 0.84  |
|                             | 2                      | 0.85                | 0.81 | 0.85 | 0.81 | 0.81 | 1.35                | 1.46 | 1.05 | 1.3  | 1.35 | 1.41                | 1.41  | 1.39 | 1.44 | 1.42  | 1.69                | 1.69  | 1.65  | 1.69  | 1.69  |
|                             | 3                      | 1.45                | 1.42 | 1.45 | 1.4  | 1.4  | 1.99                | 2.15 | 1.73 | 1.98 | 2.05 | 2.64                | 2.64  | 2.6  | 2.69 | 2.65  | 2.71                | 2.67  | 2.63  | 2.72  | 2.67  |
|                             | 4                      | 2.09                | 1.98 | 1.98 | 2    | 2    | 2.65                | 2.82 | 2.28 | 2.48 | 2.6  | 3.75                | 3.76  | 3.69 | 3.84 | 3.75  | 3.62                | 3.59  | 3.56  | 3.66  | 3.56  |
|                             | 5                      | 3.04                | 2.91 | 2.81 | 2.91 | 2.95 | 3.65                | 3.72 | 3.24 | 3.4  | 3.52 | 3.88                | 3.89  | 3.79 | 3.97 | 3.88  | 3.64                | 3.74  | 3.66  | 3.72  | 3.66  |
| Camera                      | 1                      | 0.53                | 0.51 | 0.53 | 0.51 | 0.54 | 0.5                 | 0.52 | 0.53 | 0.46 | 0.55 | 1.84                | 1.65  | 1.64 | 1.72 | 1.81  | 1.85                | 1.74  | 1.84  | 1.67  | 1.81  |
|                             | 2                      | 1.21                | 1.15 | 1.22 | 1.23 | 1.21 | 1.12                | 1.15 | 1.13 | 1.04 | 1.14 | 3.46                | 3.27  | 3.15 | 3.26 | 3.64  | 3.6                 | 3.25  | 3.77  | 3.3   | 3.59  |
|                             | 3                      | 1.76                | 1.72 | 1.77 | 1.87 | 1.81 | 1.75                | 1.78 | 1.73 | 1.63 | 1.77 | 4.97                | 4.73  | 4.34 | 4.33 | 5.09  | 5.51                | 4.88  | 6.16  | 5.03  | 5.73  |
|                             | 4                      | 2.43                | 2.32 | 2.38 | 2.52 | 2.51 | 2.48                | 2.34 | 2.4  | 2.22 | 2.51 | 7.41                | 7.24  | 6.23 | 6.29 | 7.47  | 8.84                | 7.83  | 9.8   | 7.93  | 9.32  |
|                             | 5                      | 3.4                 | 3.13 | 3.32 | 3.48 | 3.48 | 3.37                | 3.22 | 3.26 | 2.95 | 3.42 | 11.21               | 10.65 | 9.62 | 9.35 | 11.25 | 13.15               | 12.08 | 13.97 | 11.71 | 13.61 |

Scenario: Lane Change in front of the Lead car on US101 Highway

| Trajectory Prediction Model |                        | CS-LSTM             |      |      |      |      |                     |      |      |      |      | STDAN               |       |      |       |      |                     |       |       |       |       |
|-----------------------------|------------------------|---------------------|------|------|------|------|---------------------|------|------|------|------|---------------------|-------|------|-------|------|---------------------|-------|-------|-------|-------|
| Manouver                    |                        | Right Lane change   |      |      |      |      | Left Lane change    |      |      |      |      | Right Lane change   |       |      |       |      | Left Lane change    |       |       |       |       |
| Sensor Parameter            |                        | RMSE (m) at 'Range' |      |      |      |      | RMSE (m) at 'Range' |      |      |      |      | RMSE (m) at 'Range' |       |      |       |      | RMSE (m) at 'Range' |       |       |       |       |
| Sensor Modality             | Prediction Horizon (s) | 50m                 | 75m  | 100m | 125m | 150m | 50m                 | 75m  | 100m | 125m | 150m | 50m                 | 75m   | 100m | 125m  | 150m | 50m                 | 75m   | 100m  | 125m  | 150m  |
| Radar                       | 1                      | 0.45                | 0.53 | 0.51 | 0.61 | 0.48 | 0.89                | 0.81 | 0.87 | 0.77 | 0.92 | 0.77                | 0.78  | 0.78 | 0.77  | 0.77 | 0.88                | 0.85  | 0.86  | 0.91  | 0.87  |
|                             | 2                      | 1.03                | 1.16 | 1.13 | 1.08 | 1.01 | 1.83                | 1.47 | 1.64 | 1.44 | 1.86 | 1.59                | 1.64  | 1.65 | 1.64  | 1.65 | 1.78                | 1.72  | 1.73  | 1.85  | 1.76  |
|                             | 3                      | 1.55                | 1.69 | 1.67 | 1.73 | 1.64 | 2.58                | 2.15 | 2.31 | 2.23 | 2.71 | 2.67                | 2.79  | 2.79 | 2.79  | 2.8  | 2.99                | 2.88  | 2.9   | 3.06  | 2.95  |
|                             | 4                      | 2.07                | 2.17 | 2.19 | 2.26 | 2.29 | 3.41                | 2.73 | 2.86 | 3.26 | 3.45 | 3.88                | 4.04  | 4.06 | 4.07  | 4.05 | 4.31                | 4.18  | 4.2   | 4.44  | 4.25  |
|                             | 5                      | 3                   | 3.18 | 3.2  | 3.27 | 3.45 | 4.39                | 3.69 | 3.76 | 4.46 | 4.3  | 4.23                | 4.3   | 4.37 | 4.36  | 4.37 | 4.69                | 4.48  | 4.51  | 4.98  | 4.57  |
| Camera                      | 1                      | 0.95                | 0.96 | 0.96 | 0.99 | 0.94 | 0.99                | 0.99 | 0.99 | 0.98 | 0.97 | 1.73                | 1.88  | 1.47 | 1.72  | 1.71 | 2.21                | 2.52  | 2.31  | 2.3   | 1.99  |
|                             | 2                      | 1.89                | 1.93 | 1.93 | 1.99 | 1.92 | 2                   | 1.96 | 1.98 | 1.96 | 1.95 | 3.23                | 3.71  | 2.72 | 3.2   | 3.27 | 4.15                | 5.55  | 4.67  | 4.95  | 4.34  |
|                             | 3                      | 2.85                | 2.86 | 2.92 | 2.96 | 2.85 | 3                   | 2.93 | 2.92 | 2.89 | 2.87 | 4.62                | 5.35  | 4.13 | 4.73  | 4.87 | 6.22                | 8.05  | 7.12  | 7.37  | 7     |
|                             | 4                      | 3.77                | 3.81 | 3.85 | 3.93 | 3.81 | 4.04                | 3.93 | 3.92 | 3.87 | 3.84 | 7                   | 8.12  | 6.13 | 7.2   | 6.86 | 9.17                | 11.12 | 10.43 | 10.25 | 10.71 |
|                             | 5                      | 5.04                | 4.94 | 5.11 | 5.11 | 4.82 | 5.29                | 5.08 | 5.08 | 5    | 4.99 | 9.74                | 11.88 | 9.41 | 10.91 | 9.81 | 12.33               | 13.43 | 13.21 | 12.36 | 13.89 |

Scenario: Deceleration on US101 Highway

| Trajectory Prediction Model |                        | CS-LSTM             |      |      |      |      | STDAN               |      |      |      |      |
|-----------------------------|------------------------|---------------------|------|------|------|------|---------------------|------|------|------|------|
| Manouver                    |                        | Deceleration        |      |      |      |      | Deceleration        |      |      |      |      |
| Sensor Parameter            |                        | RMSE (m) at 'Range' |      |      |      |      | RMSE (m) at 'Range' |      |      |      |      |
| Sensor Modality             | Prediction Horizon (s) | 50m                 | 75m  | 100m | 125m | 150m | 50m                 | 75m  | 100m | 125m | 150m |
| Radar                       | 1                      | 0.28                | 0.29 | 0.31 | 0.33 | 0.28 | 0.31                | 0.32 | 0.32 | 0.32 | 0.31 |
|                             | 2                      | 0.65                | 0.63 | 0.7  | 0.75 | 0.61 | 1.01                | 1.02 | 1.02 | 1.02 | 1.02 |
|                             | 3                      | 1.27                | 1.24 | 1.32 | 1.39 | 1.23 | 1.9                 | 1.92 | 1.93 | 1.92 | 1.92 |
|                             | 4                      | 1.68                | 1.65 | 1.77 | 1.9  | 1.62 | 2.79                | 2.81 | 2.82 | 2.81 | 2.81 |
|                             | 5                      | 2.49                | 2.44 | 2.56 | 2.71 | 2.39 | 2.92                | 2.95 | 2.96 | 2.94 | 2.95 |
| Camera                      | 1                      | 0.37                | 0.4  | 0.38 | 0.38 | 0.41 | 1.27                | 1.16 | 0.98 | 1.03 | 1.21 |
|                             | 2                      | 0.84                | 0.92 | 0.88 | 0.82 | 0.93 | 2.39                | 2.1  | 1.85 | 1.79 | 2.3  |
|                             | 3                      | 1.47                | 1.55 | 1.53 | 1.45 | 1.59 | 3.52                | 3.02 | 2.59 | 2.51 | 3.25 |
|                             | 4                      | 2.03                | 2.12 | 2.1  | 2.01 | 2.21 | 5.01                | 4.25 | 3.67 | 3.49 | 4.64 |
|                             | 5                      | 2.95                | 3.15 | 3.02 | 2.93 | 3.17 | 7.28                | 6.57 | 5.67 | 5.32 | 6.76 |

## Sensor Modality Selection – HFOV

Scenario: Lane Change in front of the Ego car on US101 Highway

| Trajectory Prediction Model |                        | CS-LSTM            |       |       |        |        |        |                    |       |       |        |        |        | STDAN              |       |       |        |        |        |                    |       |       |        |        |        |
|-----------------------------|------------------------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|
| Manouver                    |                        | Right Lane change  |       |       |        |        |        | Left Lane change   |       |       |        |        |        | Right Lane change  |       |       |        |        |        | Left Lane change   |       |       |        |        |        |
| Sensor Parameter            |                        | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        |
| Sensor Modality             | Prediction Horizon (s) | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg |
| Radar                       | 1                      | 0.31               | 0.53  | 0.37  | 0.39   | 0.38   | 0.37   | 0.34               | 0.33  | 0.49  | 0.87   | 0.62   | 0.52   | 0.35               | 0.55  | 0.48  | 0.52   | 0.56   | 0.52   | 0.39               | 0.38  | 0.82  | 0.85   | 0.84   | 0.75   |
|                             | 2                      | 0.7                | 0.98  | 0.83  | 0.87   | 0.83   | 0.82   | 0.75               | 0.75  | 1.08  | 1.87   | 1.25   | 1.04   | 1.13               | 1.48  | 1.35  | 1.42   | 1.46   | 1.4    | 0.99               | 0.98  | 1.68  | 1.71   | 1.69   | 1.57   |
|                             | 3                      | 1.25               | 1.7   | 1.43  | 1.48   | 1.45   | 1.41   | 1.32               | 1.31  | 1.69  | 2.8    | 1.94   | 1.54   | 2.34               | 2.74  | 2.57  | 2.67   | 2.67   | 2.62   | 1.75               | 1.73  | 2.68  | 2.75   | 2.69   | 2.54   |
|                             | 4                      | 1.62               | 2.39  | 1.99  | 2.11   | 2.05   | 2.01   | 1.79               | 1.8   | 2.25  | 3.63   | 2.47   | 2.11   | 3.37               | 3.92  | 3.69  | 3.8    | 3.77   | 3.72   | 2.42               | 2.38  | 3.63  | 3.7    | 3.62   | 3.44   |
|                             | 5                      | 2.32               | 3.42  | 2.87  | 3.02   | 2.99   | 2.98   | 2.58               | 2.61  | 3.14  | 4.72   | 3.43   | 3      | 3.54               | 4.04  | 3.85  | 3.93   | 3.84   | 3.83   | 2.87               | 2.91  | 3.7   | 3.74   | 3.66   | 3.51   |
| Camera                      | 1                      | 0.55               | 0.56  | 0.55  | 0.53   | 0.52   | 0.51   | 0.59               | 0.53  | 0.53  | 0.52   | 0.54   | 0.52   | 1.93               | 1.6   | 1.64  | 1.65   | 1.74   | 1.66   | 1.86               | 1.95  | 1.8   | 1.79   | 1.94   | 1.8    |
|                             | 2                      | 1.3                | 1.19  | 1.21  | 1.17   | 1.18   | 1.19   | 1.27               | 1.2   | 1.11  | 1.16   | 1.15   | 1.16   | 3.67               | 2.98  | 3.03  | 3.3    | 3.5    | 3.25   | 3.53               | 3.88  | 3.53  | 3.63   | 3.76   | 3.52   |
|                             | 3                      | 1.87               | 1.75  | 1.77  | 1.71   | 1.76   | 1.79   | 1.89               | 1.75  | 1.73  | 1.74   | 1.73   | 1.77   | 5.15               | 4.42  | 4.34  | 4.49   | 4.97   | 4.57   | 5.25               | 5.99  | 5.47  | 5.79   | 5.63   | 5.41   |
|                             | 4                      | 2.6                | 2.41  | 2.41  | 2.35   | 2.43   | 2.47   | 2.65               | 2.46  | 2.45  | 2.49   | 2.45   | 2.53   | 7.53               | 6.67  | 6.55  | 6.82   | 7.05   | 7      | 8.69               | 9.45  | 8.76  | 8.85   | 8.92   | 8.79   |
|                             | 5                      | 3.56               | 3.41  | 3.31  | 3.16   | 3.24   | 3.37   | 3.67               | 3.28  | 3.24  | 3.43   | 3.43   | 3.42   | 10.71              | 9.98  | 10.06 | 10.49  | 10.54  | 10.67  | 13.44              | 13.62 | 13.05 | 12.76  | 13.37  | 13.14  |

Scenario: Lane Change in front of the Lead Car on US101 Highway

| Trajectory Prediction Model |                        | CS-LSTM            |       |       |        |        |        |                    |       |       |        |        |        | STDAN              |       |       |        |        |        |                    |       |       |        |        |        |
|-----------------------------|------------------------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|
| Manouver                    |                        | Right Lane change  |       |       |        |        |        | Left Lane change   |       |       |        |        |        | Right Lane change  |       |       |        |        |        | Left Lane change   |       |       |        |        |        |
| Sensor Parameter            |                        | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        |
| Sensor Modality             | Prediction Horizon (s) | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg |
| Radar                       | 1                      | 0.36               | 0.88  | 0.45  | 0.43   | 0.47   | 0.82   | 0.37               | 0.38  | 0.77  | 0.9    | 0.91   | 0.97   | 0.4                | 0.79  | 0.78  | 0.75   | 0.76   | 0.77   | 0.45               | 0.49  | 0.85  | 0.87   | 0.85   | 0.89   |
|                             | 2                      | 0.81               | 1.74  | 1.02  | 0.99   | 1.02   | 1.61   | 0.82               | 0.84  | 1.41  | 1.81   | 1.77   | 1.81   | 0.9                | 1.7   | 1.65  | 1.6    | 1.63   | 1.64   | 0.99               | 1.04  | 1.73  | 1.75   | 1.73   | 1.78   |
|                             | 3                      | 1.36               | 2.52  | 1.54  | 1.6    | 1.68   | 2.25   | 1.36               | 1.36  | 2.1   | 2.51   | 2.49   | 2.64   | 1.83               | 2.87  | 2.78  | 2.72   | 2.78   | 2.79   | 1.7                | 1.75  | 2.91  | 2.95   | 2.91   | 2.93   |
|                             | 4                      | 1.76               | 3.04  | 2.07  | 2.27   | 2.34   | 2.9    | 1.85               | 1.93  | 2.67  | 3.21   | 3.23   | 3.31   | 2.76               | 4.18  | 4.03  | 3.92   | 4.04   | 4.09   | 2.43               | 2.5   | 4.22  | 4.25   | 4.25   | 4.2    |
|                             | 5                      | 2.45               | 4.11  | 3.02  | 3.31   | 3.51   | 3.88   | 2.62               | 2.64  | 3.66  | 4.08   | 4.13   | 4.22   | 3.14               | 4.42  | 4.31  | 4.25   | 4.37   | 4.37   | 3.13               | 3.27  | 4.55  | 4.55   | 4.58   | 4.57   |
| Camera                      | 1                      | 1.13               | 1.01  | 0.99  | 0.97   | 1.01   | 0.97   | 1.14               | 1.02  | 0.98  | 0.97   | 1      | 0.93   | 1.79               | 1.56  | 1.62  | 1.4    | 1.74   | 1.69   | 2.4                | 3.06  | 2.25  | 2.15   | 2.25   | 2.16   |
|                             | 2                      | 2.26               | 2.04  | 1.99  | 1.94   | 2.03   | 1.95   | 2.26               | 2.03  | 1.93  | 1.92   | 2.01   | 1.85   | 3.44               | 3.18  | 3.15  | 2.69   | 3.41   | 3.2    | 5.22               | 6.32  | 4.26  | 4.39   | 4.06   | 4.55   |
|                             | 3                      | 3.33               | 3.06  | 2.99  | 2.86   | 3.02   | 2.91   | 3.33               | 3.01  | 2.91  | 2.84   | 2.99   | 2.79   | 5.14               | 4.85  | 4.56  | 4.04   | 4.96   | 4.85   | 8.59               | 8.97  | 6.34  | 6.92   | 5.69   | 7.3    |
|                             | 4                      | 4.44               | 4.09  | 3.95  | 3.79   | 4.03   | 3.86   | 4.44               | 4.01  | 3.87  | 3.84   | 4.05   | 3.73   | 7.55               | 7.3   | 6.99  | 6.1    | 7.1    | 7.06   | 12.85              | 11.85 | 9.39  | 10.17  | 8.97   | 10.56  |
|                             | 5                      | 5.77               | 5.37  | 5.11  | 4.9    | 5.29   | 5.06   | 5.72               | 5.37  | 5.02  | 5.07   | 5.3    | 4.81   | 11.19              | 10.39 | 10.54 | 9.72   | 10.75  | 10.37  | 16.11              | 13.6  | 12.45 | 13.08  | 12.78  | 13.54  |

Scenario: Deceleration on US101 Highway

| Trajectory Prediction Model |                       | CS-LSTM            |       |       |        |        |        | STDAN              |       |       |        |        |        |
|-----------------------------|-----------------------|--------------------|-------|-------|--------|--------|--------|--------------------|-------|-------|--------|--------|--------|
| Manouver                    |                       | Deceleration       |       |       |        |        |        | Deceleration       |       |       |        |        |        |
| Sensor Parameter            |                       | RMSE (m) at 'HFOV' |       |       |        |        |        | RMSE (m) at 'HFOV' |       |       |        |        |        |
| Sensor Modality             | Prediction Horizon(s) | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg | 30deg              | 60deg | 90deg | 120deg | 150deg | 180deg |
| Radar                       | 1                     | 0.31               | 0.3   | 0.32  | 0.3    | 0.3    | 0.27   | 0.24               | 0.24  | 0.31  | 0.31   | 0.31   | 0.32   |
|                             | 2                     | 0.69               | 0.66  | 0.69  | 0.68   | 0.67   | 0.61   | 0.76               | 0.76  | 1.02  | 1.02   | 1.01   | 1.02   |
|                             | 3                     | 1.33               | 1.28  | 1.3   | 1.29   | 1.28   | 1.23   | 1.56               | 1.56  | 1.92  | 1.92   | 1.92   | 1.92   |
|                             | 4                     | 1.78               | 1.71  | 1.76  | 1.7    | 1.7    | 1.61   | 2.38               | 2.37  | 2.81  | 2.81   | 2.81   | 2.82   |
|                             | 5                     | 2.63               | 2.53  | 2.59  | 2.52   | 2.51   | 2.41   | 2.4                | 2.38  | 2.95  | 2.95   | 2.95   | 2.96   |
| Camera                      | 1                     | 0.36               | 0.4   | 0.39  | 0.42   | 0.4    | 0.38   | 1.06               | 1.14  | 1.26  | 1.07   | 1.12   | 1.26   |
|                             | 2                     | 0.84               | 0.9   | 0.86  | 0.97   | 0.9    | 0.87   | 2.06               | 2.18  | 2.3   | 2.03   | 2.07   | 2.28   |
|                             | 3                     | 1.49               | 1.53  | 1.5   | 1.62   | 1.54   | 1.52   | 3.05               | 3.17  | 3.21  | 2.78   | 2.83   | 3.28   |
|                             | 4                     | 2.01               | 2.11  | 2.06  | 2.21   | 2.1    | 2.08   | 4.44               | 4.55  | 4.48  | 3.88   | 4.07   | 4.52   |
|                             | 5                     | 2.95               | 3.09  | 3.05  | 3.25   | 3.12   | 3.01   | 6.83               | 6.92  | 6.77  | 5.95   | 6.19   | 6.65   |